

Anakin Dey

Curriculum Vitæ

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Research Interests

Algebraic and Geometric Complexity Theory, Algebraic Combinatorics, Computational Algebra

Education

- Autumn 2024 – **PhD Mathematics**, *The Ohio State University*
Spring 2030 Advised by Dave Anderson and Zeyu Guo
- Autumn 2020 – **BSLAS Mathematics**, *University of Illinois Urbana-Champaign*
Spring 2024 Minor in Computer Science
Graduation with Highest Distinction

Publications

- 2025 **Debordering Closure Results in Determinantal and Pfaffian Ideals**
Anakin Dey and Zeyu Guo
Innovations in Theoretical Computer Science 2026, [LIPICs](#)
- 2025 **Optimizing a Model-Agnostic Measure of Graph Counterdeceptiveness via Reattachment**
Anakin Dey, Sam Ruggerio, Manav Vora, and Melkior Ornik
IEEE Cyber Awareness and Research Symposium 2025, [IEEE](#)
- 2024 **Classifying Primitive Solvable Permutation Groups of Rank 5 and 6**
Anakin Dey, Kolton O'Neal, Duc Van Khanh Tran, Camron Upshur, and Yong Yang
Journal of Group Theory, [Degruyter](#)
- 2022 **Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency**
Anakin Dey and Melkior Ornik
IEEE International Smart Cities Conference 2022, [IEEE](#)

Talks

- 2026 **What Is Hilbert's 17th Problem?**
OSU What Is...? Seminar
- 2026 **Debordering Closure Results in Determinantal and Pfaffian Ideals**
9th Workshop on Algebraic Complexity Theory
- 2026 **Symmetric Functions and the Affine Symmetric Group**
RTG 2026 Workshop: Affine Combinatorics and Quantum K-Theory
- 2026 **Debordering Closure Results in Determinantal and Pfaffian Ideals**
UIUC Theory Seminar
- 2026 **Debordering Closure Results in Determinantal and Pfaffian Ideals**
Purdue Theory CS/Math Seminar
- 2026 **Debordering Closure Results in Determinantal and Pfaffian Ideals**
Innovations in Theoretical Computer Science 2026
- 2025 **The Affine Symmetric Group**
OSU Affine Grassmannian Seminar
- 2025 **What Is Algebraic Complexity Theory?**
OSU What Is...? Seminar

- 2024 **Classifying Primitive Groups using Computer Algebra and Change Making**
China-US Group Theory Summit 2024
- 2023 **Classifying Primitive Groups using Computer Algebra and Change Making**
UIUC Department of Mathematics Undergraduate Friday Seminars
- 2023 **Classifying Primitive Solvable Permutation Groups of Rank 5 and 6**
China Three Gorges University Algebra and Combinatorics Seminar
- 2022 **Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency**
IEEE International Smart Cities Conference 2022

Research Experience

- Spring 2024 **Undergraduate Research Assistant, [ICLUE](#)**
Under Prof. Alexander Yong, University of Illinois Urbana-Champaign
- Summer 2023 **Undergraduate Research Assistant, [REU in Algebra, Combinatorics, and Statistics](#)**
Under Prof. Yong Yang, Texas State University
- Autumn 2021 – **Undergraduate Research Assistant, [LEADCAT Lab](#)**
Autumn 2023 Under Prof. Melkior Ornik, University of Illinois Urbana-Champaign

Student Organizations

- Spring 2026 – **Graduate Representative, OSU [Cycle](#)**
Current Ohio State University Cycle
- Spring 2025 – **Vice President, OSU [MGSA](#)**
Current Ohio State University Math Graduate Student Association
- Autumn 2022 – **Founder and President, ACM [SIGma](#)**
Spring 2024 Special Interest Group in Math and Algorithms
- Autumn 2021 – **Admin, ACM [SIGPwny](#)**
Spring 2024 Special Interest Group in Cybersecurity

Teaching Experience

- Autumn 2025 – **Graduate Teaching Associate, The Ohio State University**
Current
 - Math 3345 – Foundations of Higher Mathematics (Spring 2026, Grader)
 - Math 1151 – Calculus I (Autumn 2025, TA)
- Spring 2021 – **Undergraduate Course Assistant, University of Illinois Urbana-Champaign**
Spring 2024
 - CS 225H – Honors Data Structures (Spring 2024)
 - CS 374 – Introduction to Algorithms and Models of Computation (Spring 2022 – Autumn 2022, Autumn 2023)
 - CS 173 – Discrete Structures (Spring 2021 – Spring 2024)
 - CS 475 – Models of Computation (Spring 2023)
 - CS 125 – Introduction to Computer Science (Spring 2021)

Mentoring

- Autumn 2026 – **Mentor, OSU [Cycle](#)**
Current
 - How Hard is it to Compute Polynomials? (Autumn 2026 – Spring 2027, 2 students)

Organization

Autumn 2026 **STEM Puzzlehunt**
Spring 2026 **Cycle/MGSA/Spectra Puzzlehunt**
Spring 2026 **5th Annual OSU Cycle Conference**
Spring 2026 **Commutative Algebra Reading Group**
Spring 2025 – Current **MGSA Seminar**

Fellowships and Awards

Autumn 2026 **David Goss Technology & Academic Innovation Stimulus Fund**
The Ohio State University Department of Mathematics
Autumn 2024 – Spring 2025 **University Fellowship**
The Ohio State University Graduate School

Posters

2026 **Debordering Closure Results in Determinantal and Pfaffian Ideals**
AlGeCom XXVII
2026 **Debordering Closure Results in Determinantal and Pfaffian Ideals**
OSU Computer Science and Engineering Poster Expo
2023 **Classifying Primitive Solvable Permutation Groups of Rank 5**
TXST REU Poster Session

Service

Autumn 2026 – Current **Steering Committee**
[Combinatorics Online](#)
Autumn 2026 – Current **Graduate Student Representative**
The Ohio State University Department of Mathematics Graduate Studies Committee
Reviewer
○ FOCS 2026

Professional Experience

Summer 2022 **Software Engineering Intern, CME Group**
Summer 2021 **Software Engineering Intern, SiteIQ**